

Part 1 :

Throughout the presidential election Democrats have generally raised more campaign funds compared to the Republicans. This trend was noticeable in years 2008 and 2020, when the Democrats were not in control of the White House, compared to 2012, large numbers of candidates have participated in Democratic presidential primary.

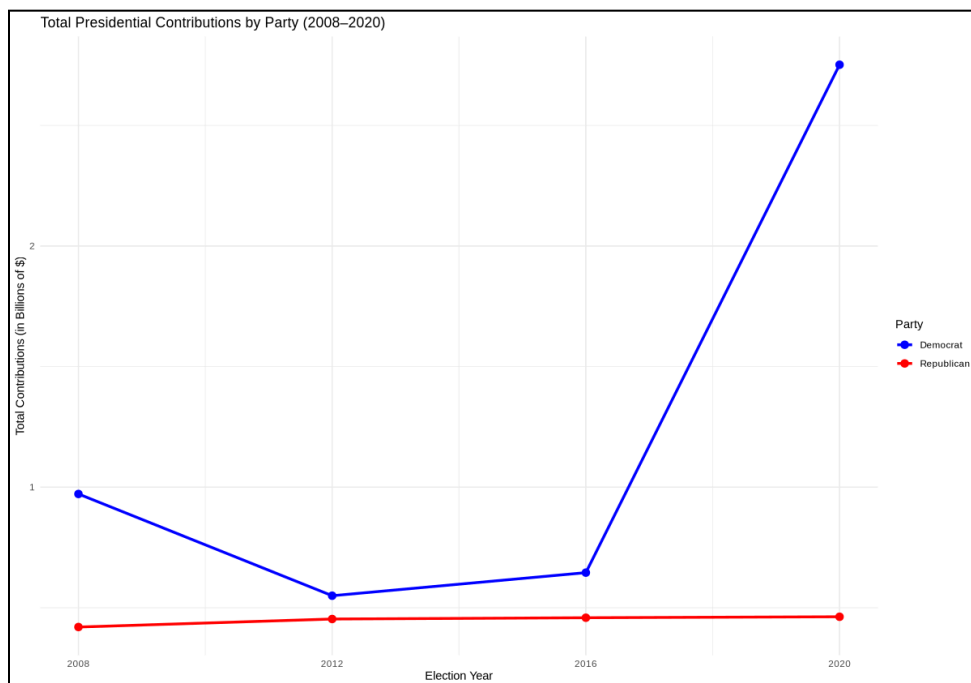
Comparing the contribution performance of the Democratic party, it is important to notice the inflation between 2008 and 2020, except for 2009, there was an annual inflation rate increase of about 1~2%, which can explain the difference in the numbers.

Also, the number of candidates running in each party should be noticed. Republicans had more candidates compared to the democrats which can decrease in the fund raising since there is much more competition. Funds raised for a specific candidate cannot be used by other candidates. Only exception in the 2024 election, Kamala Harris was Joe Biden's running mate so she could have utilized Biden's funds. However, this example is not allowed to other candidates.

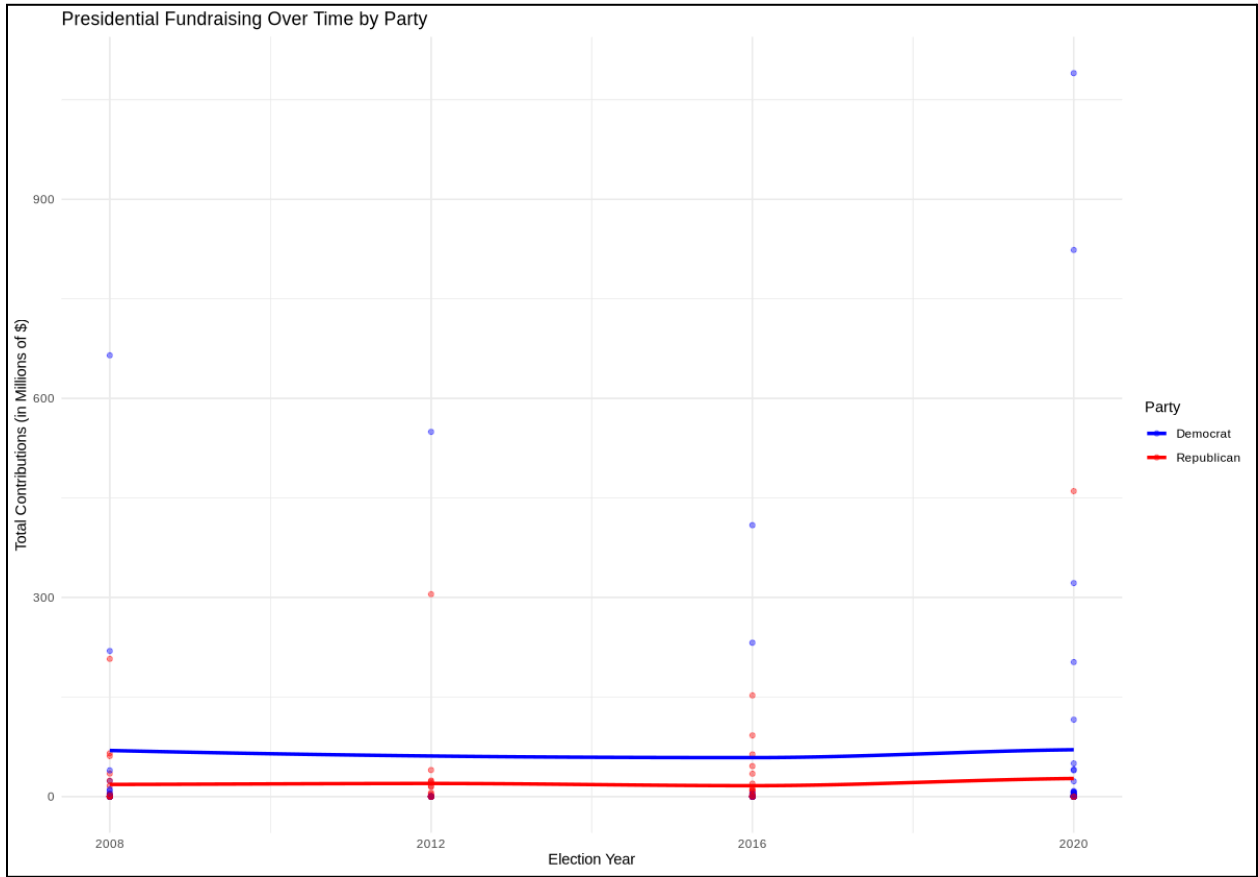
Since 2008, the Democratic Party has had a clear frontrunner from the early stages of the primaries. In contrast the Republican party has often experienced more competitive primaries, which could have resulted in lower fundraising.

In conclusion, the Democratic party have outperformed the Republican party especially in fund raising throughout 2008~2020.

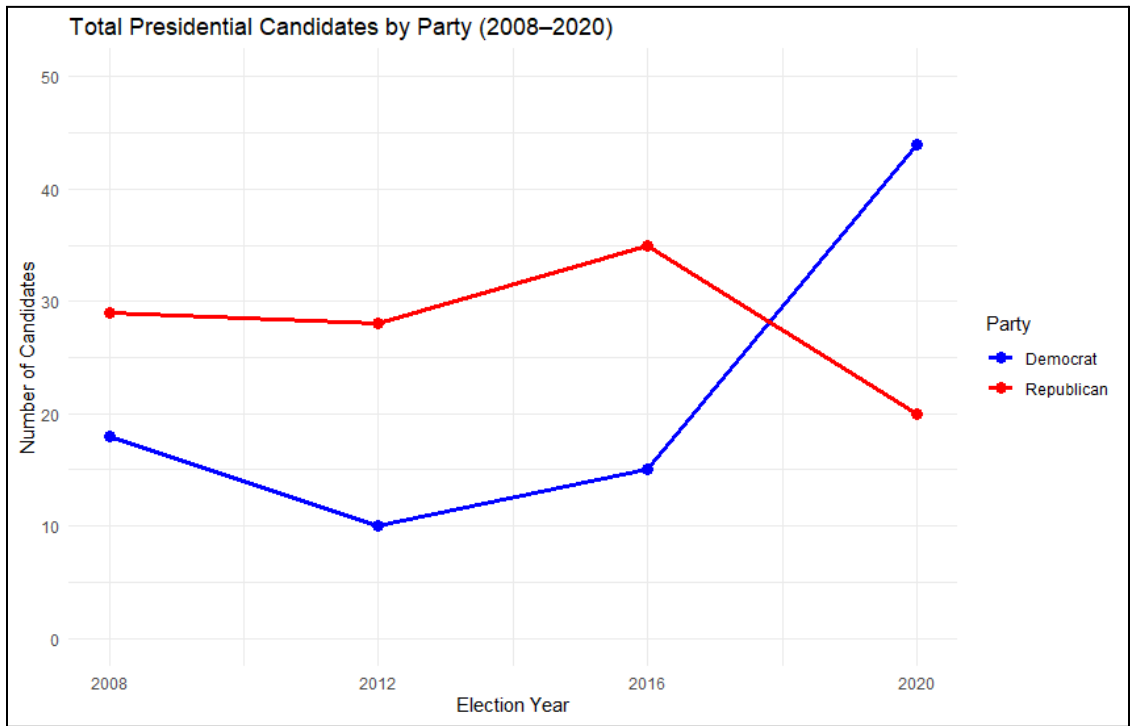
Visualization 1:



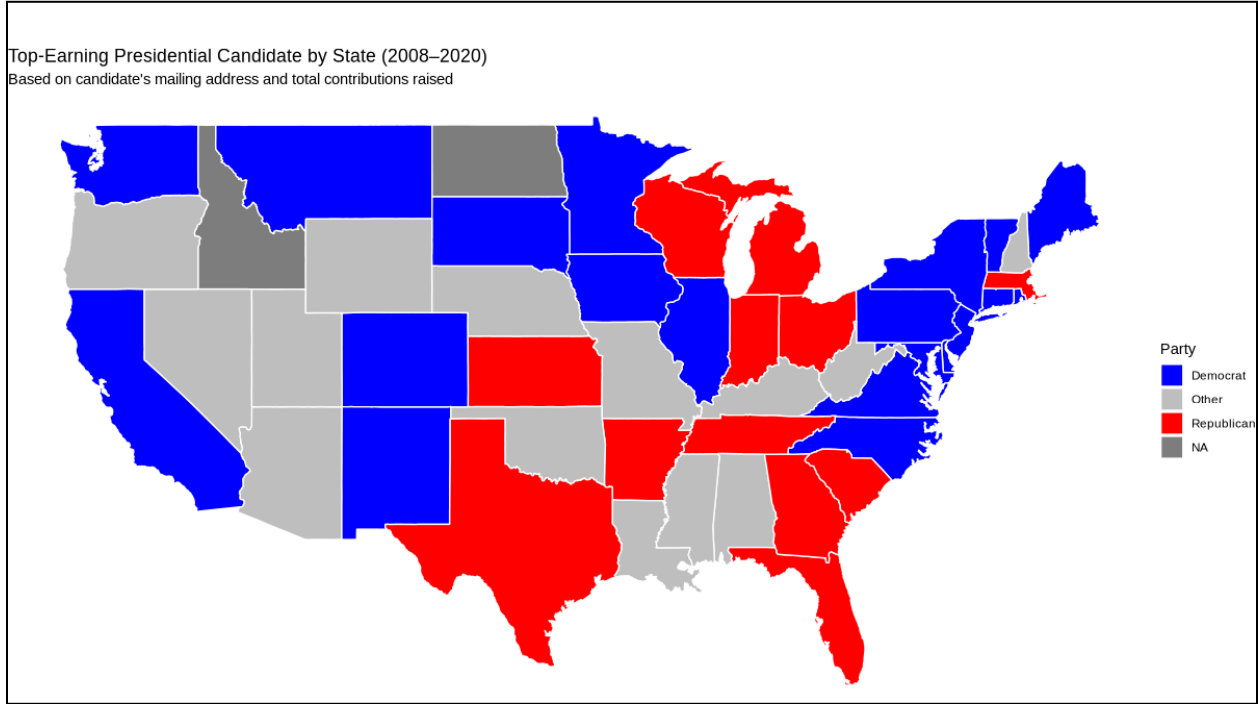
Visualization 2:



Visualization 3:



Visualization 4:

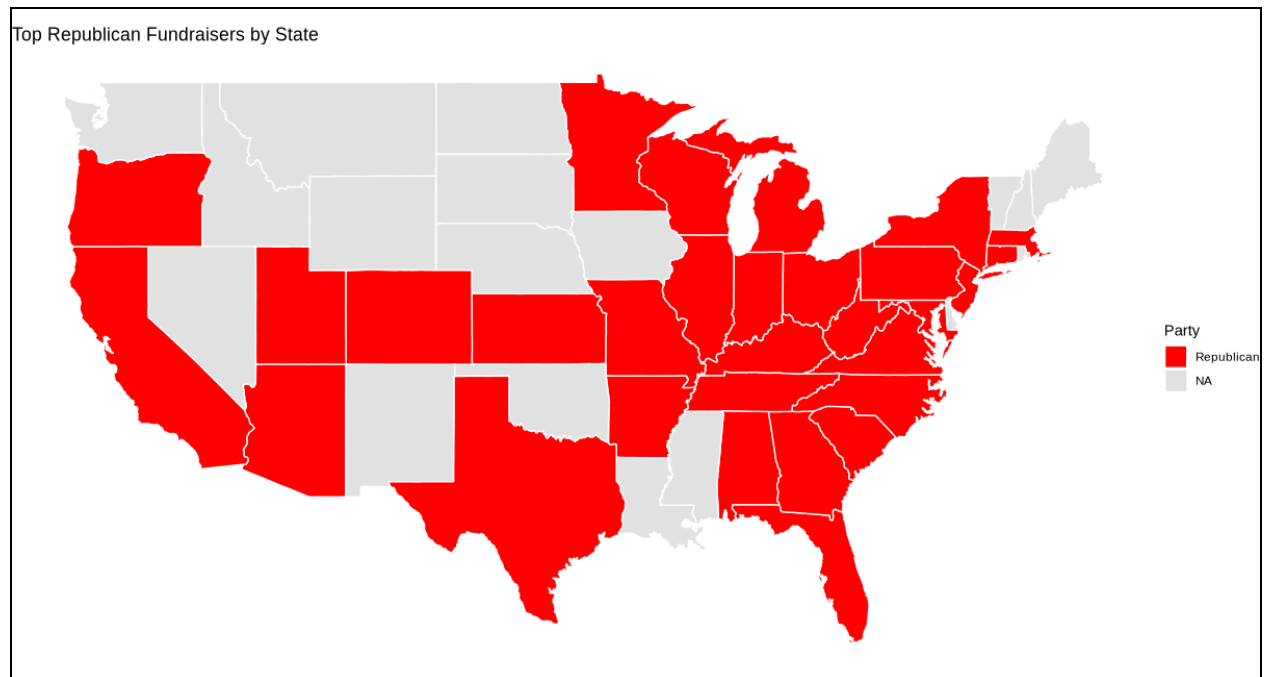


Part 2 :

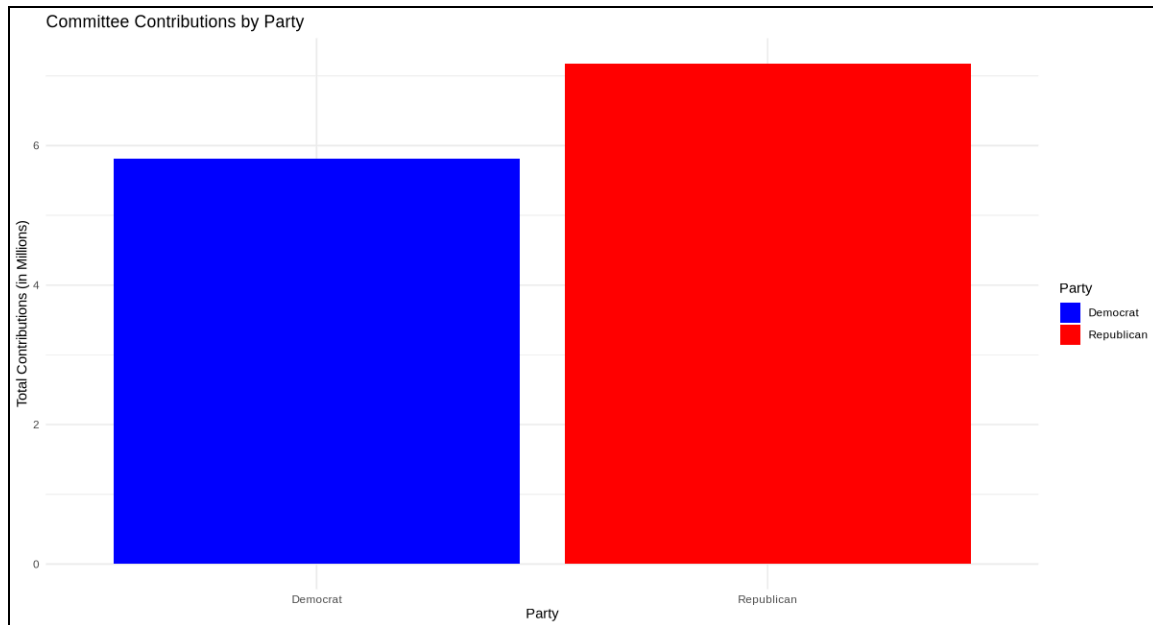
Misleading Story:

It seems that the Republican party outperforms the Democrat party in various ways in the presidential election between 2008 and 2020. Republicans had the highest contributions especially in the mid-east states. The Republican party was also able to manage much more money compared to the Democratic party. The evidence that the Republican party had much more Committee Contributions by Party, which supports the claim that the Republican party was able to manage much more money. Since, Republican party outperforms the Democratic party Republicans seems to have more candidates compared to the Democratic party for every election.

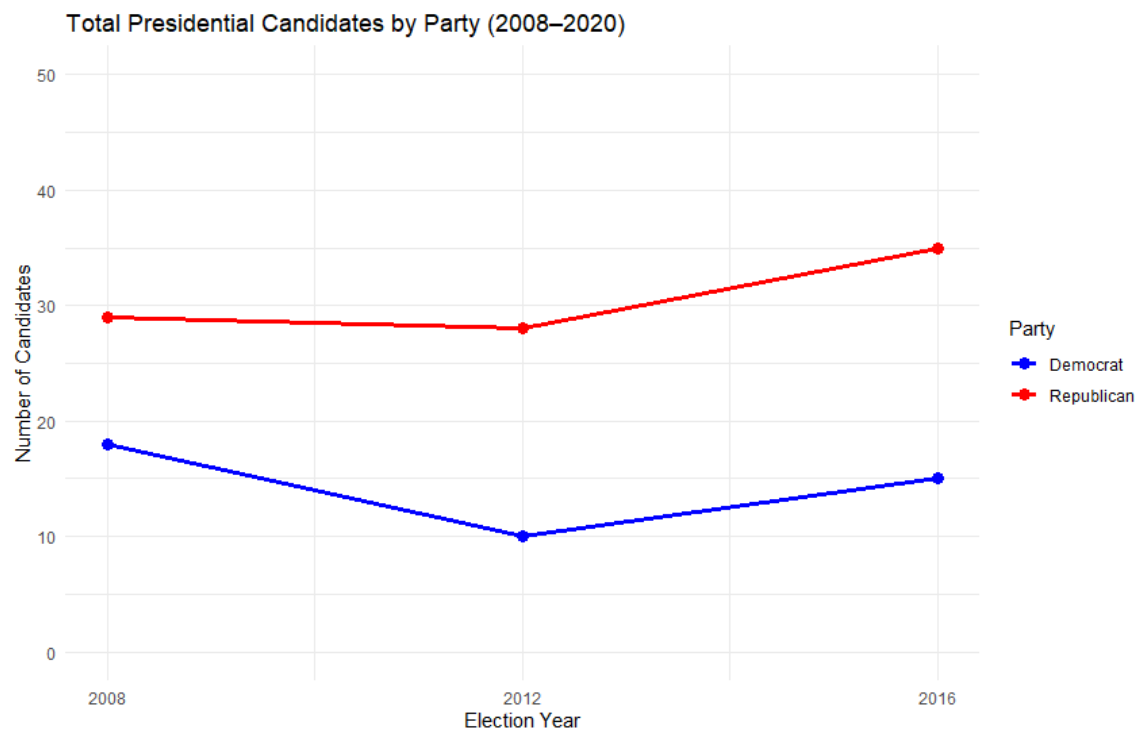
Visualization 1:



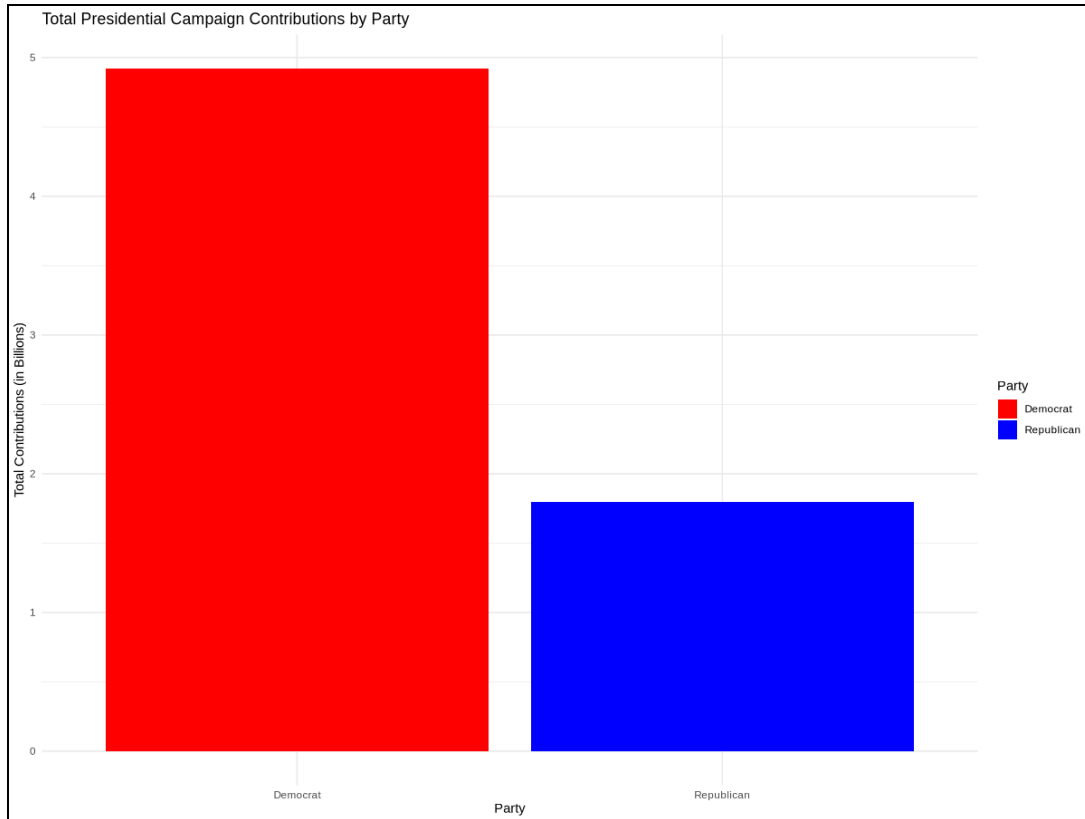
Visualization 2:



Visualization 3:



Visualization 4:



Part 3 Annotation:

The misleading visualizations can give an impression to the readers that the Republican Party outperforms the Democratic party and dominates the presidential election.

For visualization #1, the visualization only shows the states where the Republican had the highest contributions. The visualization left out that the total was not that much different from the Democratic party, this misleads the readers that Republicans have dominated.

Visualization #2 shows that the Republican party has outperformed the Democratic party in raising funds, however, the data was restricted to committee contributions by party. With respect to the total, Democratic party outperformed the Republican party, therefore, this can mislead to the readers that Republicans have outperformed in the total fund raising.

Visualization #3 also misleads the reader with limited information. In 2008, 2012, 2016 Republican party had more candidates than the Democratic party, however, in 2020 Democratic party had presidential candidates twice as much as the Republican party. But, the graph is restricted to 2008~2016 so, it seems like Republican party are always having much more candidates than the Democratic party.

Finally Visualization #4, if the readers does not carefully read the legend and the x-axis, it seems that the Republican party had outperformed the Democratic party. Since, red is the symbol of the Republican and the blue represents the Democrat. Therefore, visualizing the graph with demonstrating the Democratic color as red can mislead the readers to understand the data.

Code:

```
library(ggplot2)
library(scales)
library(dplyr)
library(maps)

a3data <- read.csv("IN4MTX 143/fec_2008-2022 (2).csv")

#####
# Visualization 1
a3data$Year <- as.numeric(format(as.Date(a3data$Coverage_End_Date, "%m/%d/%Y"),
"%Y"))

pres_data <- subset(a3data, Cand_Office == "P" & Year %in% c(2008, 2012, 2016, 2020))
pres_data$Party <- ifelse(pres_data$Cand_Party_Affiliation == "DEM", "Democrat",
                          ifelse(pres_data$Cand_Party_Affiliation == "REP", "Republican", NA))
pres_data <- subset(pres_data, !is.na(Party))

total_by_year_party <- pres_data %>%
  group_by(Year, Party) %>%
  summarise(Total = sum(Total_Contribution, na.rm = TRUE))

ggplot(total_by_year_party, aes(x = Year, y = Total / 1e9, color = Party)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Democrat" = "blue", "Republican" = "red")) +
  scale_x_continuous(breaks = c(2008, 2012, 2016, 2020)) +
  labs(title = "Total Presidential Contributions by Party (2008–2020)",
       x = "Election Year", y = "Total Contributions (in Billions of $)",
       color = "Party") +
  theme_minimal()

#####
# Visualization 2
a3data$Year <- as.numeric(format(as.Date(a3data$Coverage_End_Date, "%m/%d/%Y"),
"%Y"))
pres_data <- subset(a3data, Cand_Office == "P" & Year %in% c(2008, 2012, 2016, 2020))

pres_data$Party <- ifelse(pres_data$Cand_Party_Affiliation == "DEM", "Democrat",
                          ifelse(pres_data$Cand_Party_Affiliation == "REP", "Republican", NA))
pres_data <- subset(pres_data, !is.na(Party) & Total_Contribution > 0)
```

```
ggplot(pres_data, aes(x = Year, y = Total_Contribution / 1e6, color = Party)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE, size = 1.2) +
  scale_color_manual(values = c("Democrat" = "blue", "Republican" = "red")) +
  scale_x_continuous(breaks = c(2008, 2012, 2016, 2020)) +
  labs(title = "Presidential Fundraising Over Time by Party",
       x = "Election Year", y = "Total Contributions (in Millions of $)",
       color = "Party") +
  theme_minimal()
```

```
#####
```

```
# Visualization 3
```

```
a3data$Year <- as.numeric(format(as.Date(a3data$Coverage_End_Date, "%m/%d/%Y"),
"%Y"))
```

```
pres_data <- subset(a3data, Cand_Office == "P" & Year %in% c(2008, 2012, 2016, 2020))
pres_data$Party <- ifelse(pres_data$Cand_Party_Affiliation == "DEM", "Democrat",
                        ifelse(pres_data$Cand_Party_Affiliation == "REP", "Republican", NA))
pres_data <- subset(pres_data, !is.na(Party))
```

```
candidate_count <- pres_data %>%
  group_by(Year, Party) %>%
  summarise(Num_Candidates = n_distinct(Cand_Name), .groups = "drop")
```

```
ggplot(candidate_count, aes(x = Year, y = Num_Candidates, color = Party)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Democrat" = "blue", "Republican" = "red")) +
  scale_x_continuous(breaks = c(2008, 2012, 2016, 2020)) +
  labs(title = "Total Presidential Candidates by Party (2008–2020)",
       x = "Election Year", y = "Number of candidates",
       color = "Party") +
  theme_minimal()
```

```
#####
```

```
# Visualization 4
```

```
a3data$Year <- as.numeric(format(as.Date(a3data$Coverage_End_Date, "%m/%d/%Y"),
"%Y"))
pres_data <- a3data %>%
  filter(Cand_Office == "P", Year %in% c(2008, 2012, 2016, 2020)) %>%
```

```
mutate(
  Cand_State = toupper(trimws(Cand_State)),
  Party = case_when(
    Cand_Party_Affiliation == "DEM" ~ "Democrat",
    Cand_Party_Affiliation == "REP" ~ "Republican",
    TRUE ~ "Other"
  )
)
```

```
valid_states <- state.abb
state_map <- data.frame(Cand_State = state.abb, region = tolower(state.name),
stringsAsFactors = FALSE)
```

```
pres_data <- pres_data %>%
  filter(Cand_State %in% valid_states) %>%
  left_join(state_map, by = "Cand_State")
```

```
top_earners <- pres_data %>%
  group_by(region) %>%
  slice_max(order_by = Total_Contribution, n = 1, with_ties = FALSE) %>%
  select(region, Cand_Name, Total_Contribution, Party)
```

```
us_map <- map_data("state")
```

```
map_data <- left_join(us_map, top_earners, by = "region")
```

```
ggplot(map_data, aes(long, lat, group = group, fill = Party)) +
  geom_polygon(color = "white") +
  coord_fixed(1.3) +
  scale_fill_manual(values = c("Democrat" = "blue", "Republican" = "red", "Other" = "gray")) +
  labs(
    title = "Top-Earning Presidential Candidate by State (2008–2020)",
    subtitle = "Based on candidate's mailing address and total contributions raised",
    fill = "Party"
  ) +
  theme_void()
```

```
#####  
##### PART 2 #####  
#####
```

```
#Wrong Visualization #1  
map_data_mis <- pres_data %>%  
  filter(Party == "Republican") %>%  
  group_by(region) %>%  
  slice_max(Total_Contribution, n = 1, with_ties = FALSE)  
  
map_us <- map_data("state")  
  
map_join <- left_join(map_us, map_data_mis, by = "region")  
  
ggplot(map_join, aes(long, lat, group = group, fill = Party)) +  
  geom_polygon(color = "white") +  
  coord_fixed(1.3) +  
  scale_fill_manual(values = c("Republican" = "red"), na.value = "gray90") +  
  labs(title = "Top Republican Fundraisers by State") +  
  theme_void()
```

```
#####
```

```
#Visualization # 2  
a3data$Year <- as.numeric(format(as.Date(a3data$Coverage_End_Date, "%m/%d/%Y"),  
"%Y"))  
  
pres_data <- a3data %>%  
  filter(Cand_Office == "P", Year %in% c(2008, 2012, 2016, 2020)) %>%  
  mutate(Party = case_when(  
    grepl("(?i)dem", Cand_Party_Affiliation) ~ "Democrat",  
    grepl("(?i)rep", Cand_Party_Affiliation) ~ "Republican",  
    TRUE ~ "Other"  
  )) %>%  
  filter(Party %in% c("Democrat", "Republican"))  
  
other_committee_totals <- pres_data %>%  
  group_by(Party) %>%  
  summarise(Total_Other_Committee_Contribution = sum(Other_Committee_Contribution,  
na.rm = TRUE))  
  
ggplot(other_committee_totals, aes(x = Party, y = Total_Other_Committee_Contribution / 1e6, fill  
= Party)) +  
  geom_bar(stat = "identity") +
```

```

scale_fill_manual(values = c("Democrat" = "blue", "Republican" = "red")) +
labs(
  title = "Committee Contributions by Party ",
  x = "Party",
  y = "Total Contributions (in Millions)",
  fill = "Party"
) +
theme_minimal()

```

#####

#Wrong Visualization for #3

```

a3data$Year <- as.numeric(format(as.Date(a3data$Coverage_End_Date, "%m/%d/%Y"),
"%Y"))

```

```

pres_data <- subset(a3data, Cand_Office == "P" & Year %in% c(2008, 2012, 2016))
pres_data$Party <- ifelse(pres_data$Cand_Party_Affiliation == "DEM", "Democrat",
  ifelse(pres_data$Cand_Party_Affiliation == "REP", "Republican", NA))
pres_data <- subset(pres_data, !is.na(Party))

```

```

candidate_count <- pres_data %>%
  group_by(Year, Party) %>%
  summarise(Num_Candidates = n_distinct(Cand_Name), .groups = "drop")

```

```

ggplot(candidate_count, aes(x = Year, y = Num_Candidates, color = Party)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Democrat" = "blue", "Republican" = "red")) +
  scale_x_continuous(breaks = c(2008, 2012, 2016)) +
  labs(title = "Total Presidential Candidates by Party (2008–2020)",
    x = "Election Year", y = "Number of Candidates",
    color = "Party") +
  scale_y_continuous(limits = c(0, 50)) +
  theme_minimal()

```

#####

#Wrong Visualization 4

```

a3data$Year <- as.numeric(format(as.Date(a3data$Coverage_End_Date, "%m/%d/%Y"),
"%Y"))

```

```

pres_data <- a3data %>%
  filter(Cand_Office == "P", Year %in% c(2008, 2012, 2016, 2020)) %>%
  mutate(Party = case_when(
    grepl("(?i)dem", Cand_Party_Affiliation) ~ "Democrat",
    grepl("(?i)rep", Cand_Party_Affiliation) ~ "Republican",
    TRUE ~ "Other"
  )) %>%
  filter(Party %in% c("Democrat", "Republican"))

party_totals <- pres_data %>%
  group_by(Party) %>%
  summarise(Total_Contribution = sum(Total_Contribution, na.rm = TRUE))

ggplot(party_totals, aes(x = Party, y = Total_Contribution / 1e9, fill = Party)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("Democrat" = "red", "Republican" = "blue")) +
  labs(
    title = "Total Presidential Campaign Contributions by Party",
    x = "Party",
    y = "Total Contributions (in Billions)",
    fill = "Party"
  ) +
  theme_minimal()

```